PART A: PROJECT THEORY

Spam Message Classification Using Probability and Machine Learning

1. Introduction

Spam messages are unwanted or unsolicited digital messages sent through communication platforms such as SMS, email, and messaging applications. These messages may contain advertisements, misleading information, suspicious links, or fraudulent content. Detecting spam automatically is important for improving communication security and user experience.

Machine learning can identify patterns in message-related data and classify messages into two categories: Spam and Legitimate. This project applies probability and machine learning concepts to build and compare classification models for spam message detection.

2. Problem Statement

The objective of this project is to develop a classification system that predicts whether a digital message is spam or legitimate using numerical features associated with the message. The system will learn from labeled examples and evaluate its predictions on unseen testing data.

3. Project Objectives

- Understand and prepare a dataset containing message-related features and a spam label.
- Apply a train-test split to evaluate models on unseen observations.
- Use feature scaling where required by the algorithms.
- Implement K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naive Bayes classifiers.
- Evaluate model performance using Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.
- Compare the models and identify a suitable classifier for spam detection.

4. Dataset Description

The project uses the Message Intelligence dataset, containing 5,200 records and 16 columns, as provided for this project. The target column is spam_label. The dataset includes message-related identifiers, message text and/or engineered numerical indicators, time-related information, and features that describe message characteristics.

For the numerical-feature classification workflow, the identifier columns and raw text/timestamp fields are excluded from the model inputs where appropriate. The target variable is separated from the input features.

- Target variable: spam_label
- Class 0: Legitimate message
- Class 1: Spam message
- Example numerical feature types: message length, special-character counts, URL counts, keyword scores, sender activity, and time-based features.

5. Machine Learning Concepts

5.1 Supervised Learning

Supervised learning uses labeled examples to learn a relationship between input features and a target label. In this project, each record has a spam_label that is used to train the classifiers.

5.2 Classification

Classification is a supervised learning task in which an observation is assigned to a category. This project is a binary classification problem because the output has two classes: Spam and Legitimate.

5.3 Probability and Bayes' Theorem

Probability expresses the likelihood of an event. Bayes' Theorem updates the probability of a class after observing evidence:

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

Here, P(A|B) is the posterior probability, P(B|A) is the likelihood, P(A) is the prior probability, and P(B) is the evidence. Naive Bayes uses this principle to estimate the probability that a message belongs to a particular class.

5.4 Feature Scaling

Feature scaling transforms numerical features to comparable scales. StandardScaler standardizes features using the training data's mean and standard deviation. Scaling is especially important for KNN and SVM. The scaler is fitted on training data only to avoid data leakage.

6. Classification Algorithms

6.1 K-Nearest Neighbors (KNN)

KNN is a distance-based algorithm. It finds the K closest training observations to a new observation and predicts the class that is most common among those neighbors. Its performance depends on the selected K value and feature scaling.

6.2 Support Vector Machine (SVM)

SVM identifies a decision boundary, called a hyperplane, that separates classes while maximizing the margin between them. The RBF kernel can model non-linear boundaries. SVM is sensitive to feature scale.

6.3 Gaussian Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It assumes that features are conditionally independent given the class. Gaussian Naive Bayes models numerical feature values with Gaussian distributions and estimates the probability of each class.

7. Model Evaluation Metrics

- Accuracy: The proportion of all predictions that are correct.
- Precision: Of the messages predicted as spam, the proportion that are truly spam.
- Recall: Of all actual spam messages, the proportion correctly detected.
- F1 Score: The harmonic mean of precision and recall.
- Confusion Matrix: A table showing true positives, true negatives, false positives, and false negatives.

For spam detection, recall is useful for measuring how much spam is detected, while precision helps limit legitimate messages being incorrectly flagged as spam. F1 Score balances both measures.

8. Project Methodology

- Load the dataset and inspect its shape, columns, data types, and summary statistics.
- Check for missing values and duplicate records, then clean the data as required.
- Separate the target spam_label from the input features; exclude identifiers and unsuitable raw fields from this numerical model workflow.
- Split the data into training and testing sets, using stratification to preserve class proportions.
- Scale numerical features using a scaler fitted only on the training data.
- Train KNN, SVM, and Gaussian Naive Bayes classifiers.
- Generate predictions and evaluate each model using classification metrics and confusion matrices.
- Compare results and select a model based on the chosen metric, such as F1 Score.

9. Expected Outcome

The project is expected to demonstrate how different machine learning algorithms perform on a spam message classification task. The comparison will show that model performance depends on the data, feature representation, and algorithm assumptions. The best model should be selected using the actual evaluation results rather than assumed in advance.

10. Conclusion

This project applies supervised machine learning and probability concepts to classify messages as spam or legitimate. KNN, SVM, and Gaussian Naive Bayes provide three different approaches: distance-based, margin-based, and probabilistic classification. By preparing the dataset carefully and evaluating the models with multiple metrics, the project provides a practical understanding of classification and spam detection.